

Ashwath Balaji

804-418-9085 • ashwathbalaji04@gmail.com • [linkedin.com/in/ashwathb/](https://www.linkedin.com/in/ashwathb/) • github.com/AshwathB-debug

EDUCATION

Virginia Commonwealth University, Richmond, VA

Expected May 2026

Bachelor of Science in Computer Science, Concentration in Data Science, Minor in General Business

GPA 3.4

Relevant Coursework: Algorithm Analysis with Advanced Data Structures, Databases, Fundamentals of Software Engineering, Introduction to Data Science, Machine Learning

WORK EXPERIENCE

Software Engineer Intern, Kolabrata

Jun 2025 - Present

- Developed a Gemini API powered chatbot with Python and integrated it into the front-end, enhancing user interactions by 80%; modularized code and implemented secure API key authentication.
- Collaborating with a cross-functional team to polish and optimize Kolabrata's official website; currently learning React to refactor components, improve performance, and manage backend integration and deployment via Netlify.

Undergraduate Researcher, VCU College of Engineering

May 2025 - Sep 2025

- Collaborated with Dr. Eyuphan Bulut; developed an iOS mobile app for medical image processing using OpenCV and Python to analyze the midbrain and pons for brain disease diagnosis; conducted feature testing with medical team feedback.
- Built user interface with Flutter SDK and Dart; integrated camera controls (flashlight, gallery, retake), improving usability and user experience by 50%. Developed guided overlays reducing bottom ROI misalignment by 15%.
- Implemented real-time image processing algorithms, bringing midbrain-to-pons ratio alignment within 10% of target values on test MRI images.

Product Researcher & Co-founder, NutriPicks

Nov 2024 - Aug 2025

- Co-founded a health-focused startup; conducted market research across 15+ stores and coordinated with supplier (Farmley) to optimize strategy for Indian snack products.
- Tracked 10+ leads and pricing data on Google Sheets and Excel. Negotiated case quantities to reduce financial risk by 250%.

PROJECTS

Capstone Project, AI Agent for Scientific Visualization - Project Manager & Programmer

Aug 2025 - Present

- Serving as a Project Manager for a team of 4; translated stakeholder requirements into product backlog items with defined success criteria. Facilitating weekly standups, planning deliverables, and monitoring progress in an Agile environment.
- Contributing to programming efforts, researching and testing Hugging Face Transformer models for integration into AI agent workflow, evaluating model outputs to ensure alignment with scientific visualization goals.

Financial Data Manager, Databases, VCU

Oct 2024 - Dec 2024

- Built a normalized MySQL relational database for the magnificent 7 stocks, manually ingested and integrating data from Yahoo Finance and Alpha Vantage to make rational investment decisions.
- Prototyped ER diagrams, SQL DDL/queries using Python with SQLAlchemy for CRUD operations, doubling query efficiency

Project Codex

Aug 2024 - Oct 2024

- Built an AI-powered Python chatbot integrating Google Gemini API. Deployed via Streamlit improving user engagement by 50%. Designed secure API key authentication, and modularized code with OOP principles
- Integrated text-to-speech functionality leveraging pyttsx3 library to provide realistic interaction.

SKILLS

Programming Languages & Libraries: Python, Java, C, SQL, Dart, Pandas, NumPy, OpenCV, Flutter

Tools & Methodologies: Git, Linux Command Line, Agile, MySQL, PowerShell

ACTIVITIES/CLUBS

Investment Analyst, Student Managed Investment Portfolio, VCU

Feb 2024 - Present

- Managing a \$430K equity portfolio. Collaborated on a successful sell pitch, leveraging financial analysis generating \$3500.